AYUSH JUVEKAR

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ML / Data Scientist

PROFILE

Computer science graduate with applied ML and data-analysis experience across model evaluation, computer vision, and privacy/security-focused Python workflows.

EDUCATION

Michigan Technological University, Houghton, MI — **CGPA: 3.9/4.0**
M.S. Computer Science — Graduated Apr 24, 2026

Pune Institute of Computer Technology, Pune, India — **GPA: 8.14/10**
B.E. Computer Engineering — Jun 2024

EXPERIENCE

Teaching Assistant - Physics Department | Michigan Technological University — Sep 2024 - Apr 24, 2026
Houghton, MI

- Supervised and graded physics labs for 100+ students across four semesters, guiding experiments, explaining concepts, and providing clear feedback across five lab sections per semester.

Technical Intern | CVBDSL Lab, SCTR's Pune Institute of Computer Technology — Feb 2023 - May 2023
Pune, India

- Analyzed RSA and related encryption approaches in Python/Jupyter during a four-month academic internship, comparing implementation tradeoffs for computational cost and efficiency.

SELECTED PROJECTS

Crowd Anomaly Detection Prototype | *Python, TensorFlow, OpenCV, TensorFlow Lite, Jetson Nano*

- Led model research and data collection for a BE final-year team project on edge crowd-anomaly detection, recording training videos and evaluating CNN/TensorFlow approaches on 10K+ frames.
- Optimized the prototype with TensorFlow Lite for Jetson Nano demonstration, reaching about 92% test accuracy in controlled demo runs without claiming production deployment.

Particle Vibration Analysis | *MATLAB, Computer Vision, Object Detection, ZED 2 Camera*

- Built MATLAB computer-vision scripts for particle vibration analysis from ZED 2 camera video, using object detection workflows to track particles in test recordings.
- Delivered analysis outputs for a professor's downstream PhD research, validating scripts on sample particle videos before larger research use.

Encryption Efficiency Study | *Python, Pandas, NumPy, Jupyter Notebook, RSA*

- Analyzed RSA and related encryption approaches during a CVBDSL academic internship, comparing implementation tradeoffs for computational cost and efficiency.
- Documented findings through Python/Jupyter experiments using Pandas and NumPy as part of a four-month secure data analytics module.

TECHNICAL SKILLS

Languages	Python, SQL, C++, TypeScript
ML / Data	TensorFlow, Scikit-learn, NumPy, Pandas
Computer Vision	OpenCV, image/video preprocessing
Experimentation	Model evaluation, dataset preparation, feature analysis
Tools	Jupyter Notebook, Git, Docker
Databases	PostgreSQL, MongoDB

PUBLICATION

A. Juvekar et al., “Carbon Monoxide Concentration Monitoring System,” *ESCI 2023*, presented Mar 2023. IEEE